

Laboratory Exercise 1: The Dipole Antenna

In this laboratory you will design, simulate, fabricate, and measure a center-fed dipole antenna near 1 GHz. You will choose a wire or copper-tape construction, predict its properties analytically and using CST, and compare these predictions with measurements using a LibreVNA. The exercise is intentionally project-oriented: you are responsible for developing and justifying an effective workflow rather than following a prescribed construction recipe.

Objectives

- Connect analytical dipole theory with numerical simulation and physical measurements.
- Predict resonant length, complex input impedance, and bandwidth for a chosen conductor geometry.
- Construct a working dipole and obtain calibrated reflection-coefficient measurements.
- Investigate the effects of length and, optionally, conductor diameter or strip width.
- Evaluate feedline, calibration, reference-plane, and measurement uncertainty effects.

Available equipment and materials

LibreVNA (50 Ω), SMA coaxial cables, SMA-to-two-screw-terminal adapters, soldering and hand tools, solid copper wire of different gauges, copper tape of several widths, foam-core hobby boards, insulating supports, and basic mounting hardware. CST Microwave Studio and data-analysis tools (Python or MATLAB) are available. AI assistance is permitted for troubleshooting, scripting, and analysis, but you must independently check and explain your methods.

EXERCISE 1.1: DEVELOP AN ANALYTICAL DESIGN

Develop an initial design for a center-fed dipole near 1 GHz. Estimate the free-space wavelength, choose a starting total length and feed gap, and select the conductor geometry. Use an

appropriate thin-wire dipole model to estimate radiation resistance, clearly distinguishing the electrically short-dipole approximation from a near-half-wave dipole. Consider why the zero-reactance resonant length is generally shorter than exactly one-half wavelength.

Determine what input resistance and reactance you expect near resonance, and how these values relate to the 50 Ω VNA reference impedance. Record assumptions and predictions before beginning simulation.

LAB QUESTIONS

1. What initial dimensions would you select for a dipole resonant near 1 GHz and why?
2. What input resistance do you expect at resonance, and how does this compare with the 50 Ω impedance of the VNA?
3. How should the antenna impedance change as the length is increased or decreased?
4. How might the conductor diameter or width affect resonance and bandwidth?

Document your initial design calculations and assumptions.

EXERCISE 1.2: SIMULATE THE ANTENNA IN CST

Create a 3D high-frequency model of your chosen antenna, using an appropriate center-feed excitation and open radiation boundaries. Predict complex input impedance, S11 relative to 50 Ω, and the current distribution over a frequency interval spanning 1 GHz. Choose a suitable frequency sweep and verify that mesh and boundary settings are adequate.

Perform a parametric study of total dipole length. Investigate conductor diameter or strip width if time permits. Use the results to select a practical fabrication geometry.

Consider how the antenna is fed. An idealized excitation at the center of the dipole is a reasonable starting point, but the physical antenna will eventually be connected to a coaxial cable and an adapter. Think about which features of the actual feed arrangement may affect the measurement.

Investigate the antenna response over a range of frequencies around 1 GHz.

At a minimum, obtain:

- Input impedance, including resistance and reactance, as a function of frequency.
- Reflection coefficient (S_{11}), referenced to 50 Ω.
- Resonant frequency and impedance bandwidth.
- Surface-current distribution near resonance.

Use a parameter sweep to investigate how the antenna's length affects its resonance and input impedance. You may also vary conductor diameter or strip width.

LAB QUESTIONS

5. How closely does the simulated resonant length agree with your analytical estimate?
6. What happens to the input impedance as the dipole length changes?
7. How does the predicted bandwidth depend on conductor thickness or width?
8. What current distribution is observed along the antenna near resonance?

Select a final geometry for fabrication based on your simulations.

EXERCISE 1.3: FABRICATE THE DIPOLE

Build either a free-standing copper-wire dipole or a planar copper-tape dipole on insulating hobby board. Select and document conductor dimensions, feed gap, and attachment method. The two dipole arms must remain electrically isolated except through the measurement port. Minimize unnecessary exposed feed leads and provide mechanical strain relief. Note that we can 3D print a mount for this exercise!

Record the as-built dimensions and photograph the completed antenna. If your construction differs appreciably from the original model, update the CST geometry to represent the actual antenna.

EXERCISE 1.4: CALIBRATE AND MEASURE WITH THE LIBRE VNA

The LibreVNA measures the reflection coefficient of a device relative to a specified reference impedance, normally 50 Ω.

Before measuring your antenna, determine an appropriate calibration procedure. You should understand what is accomplished by open, short, and load calibration standards, and why calibration is normally performed over the frequency range of interest. For this exercise, we can simply use the calibration procedure that is provided by LibreVNA and included in their documentation.

Determine and execute a suitable one-port VNA calibration over the measurement band. Identify the calibration reference plane: typically the end of the SMA test cable. Consider whether the screw-terminal adapter and short feed leads can be neglected, included in CST, or characterized for de-embedding. Do not apply a de-embedding correction unless you can justify the model or measurement used to obtain it.

Measure complex S_{11}, calculate input impedance using $Z_{\text{in}} = 50(1 + S_{11})/(1 - S_{11})$, and identify the frequency where the input reactance crosses zero. Separately report the frequency of minimum $|S_{11}|$; these frequencies need not coincide. Determine an impedance-bandwidth

measure (for example, the −10 dB S_{11} band, if it exists) and export data for comparison with CST.

As a diagnostic, move the coaxial feed cable while holding the antenna fixed. Consider whether feedline common-mode current affects the measurement and whether a suitable choke would be useful.

Measurement tasks:

1. Calibrate the LibreVNA over a suitable frequency range around 1 GHz.
2. Measure the antenna reflection coefficient (S_{11}).
3. Determine the resonant frequency and input impedance.
4. Determine the antenna's impedance bandwidth using an appropriate definition.
5. Export the measured data for comparison with CST.

EXERCISE 1.5: INVESTIGATE GEOMETRY

Measure at least two different dipole lengths, using either a reversible adjustment or multiple specimens. Compare the change in resonant frequency and impedance with CST predictions. An optional extension is to compare different wire gauges, tape widths, or a wire dipole with a tape dipole. Try to change only one geometric parameter at a time.

You do not need to investigate every possibility. Choose a manageable set of comparisons that will allow you to draw meaningful conclusions.

EXERCISE 1.6: COMPARISON WITH ANTENNA THEORY

Using the current distribution of a thin dipole, estimate its radiation resistance analytically.

For a sufficiently short dipole, the radiation resistance is small and increases approximately with the square of its electrical length. Near half-wave resonance, the radiation resistance approaches the familiar value associated with a center-fed half-wave dipole.

Consider the assumptions behind these approximations and their applicability to your fabricated antenna.

Compare your analytical predictions with the simulated and measured input resistance.

Remember that the input resistance of a real antenna may include conductor loss, dielectric loss, and other effects in addition to radiation resistance. Furthermore, the impedance measured at the VNA may include contributions from the feed arrangement

LAB QUESTIONS

9. How does the analytical radiation resistance compare with the simulated input resistance?
10. How closely does the measured resistance agree with simulation?
11. What physical effects could account for differences?
12. Can the measured input resistance alone establish the antenna's radiation efficiency? Explain.

Laboratory Exercise 1: Report

Prepare a concise technical report documenting your design, simulation, fabrication, and measurement process.

Your report should include:

1. **Antenna design:** Your selected geometry, dimensions, and analytical estimates.
2. **Simulation:** CST model, impedance and S11 results, current distribution, and relevant parameter sweeps.
3. **Fabrication:** Photographs, measured dimensions, and description of the feed arrangement.
4. **Measurement:** Calibration approach, reference-plane considerations, and measured complex impedance and S11.
5. **Comparison:** Overlay plots comparing simulated and measured results, including the effects of changing antenna length.
6. **Discussion:** Explanations for agreement or disagreement between theory, simulation, and experiment.

In your discussion, address the following broader questions:

- Why is the resonant length of a practical dipole generally different from exactly one-half wavelength?
- What determines the impedance bandwidth of a dipole?
- Why might a copper-tape dipole behave differently from a round-wire dipole?
- How can the feed cable influence the measurement of a nominally balanced antenna?

- What limits the accuracy of the VNA measurement, and when would de-embedding become necessary?
- Which aspects of the antenna response can be predicted reliably with simple analytical theory, and which require numerical simulation?